Lecture Notes: Machine Learning Basics

1. Fundamental Learning Paradigms

Machine learning algorithms build mathematical models from sample data to make predictions or decisions without being explicitly programmed. In supervised learning, the training set consists of input features paired with ground-truth targets, enabling regression for continuous outputs and classification for discrete labels. Conversely, unsupervised learning discovers hidden patterns or cluster structures within unlabeled datasets using methods like k-means clustering and principal component analysis.

2. Overfitting and Regularization Techniques

Overfitting occurs when an algorithm models the noise and statistical idiosyncrasies of the training data rather than the underlying general trend, yielding high training accuracy but poor generalization to test sets. Regularization mitigates overfitting by penalizing complex hypothesis weights. L1 regularization (Lasso) adds the absolute value of weights to the cost function, promoting parameter sparsity by driving irrelevant weights to zero. L2 regularization (Ridge) adds the squared magnitude of coefficients, shrinking weights smoothly toward zero without exact elimination.

3. The Bias-Variance Tradeoff

The bias-variance tradeoff is a central theoretical dilemma in statistical learning. Bias measures erroneous assumptions made by the learning algorithm; high bias leads to underfitting where models fail to capture complex relationships. Variance measures model sensitivity to slight fluctuations in training sets; high variance causes overfitting where minor sample variations alter predicted functions significantly. The total expected generalization error equals the sum of squared bias, variance, and irreducible noise.

4. Optimization and Model Validation

Gradient descent is an iterative first-order optimization algorithm that minimizes empirical loss functions by computing partial derivatives with respect to parameters and taking steps proportional to the negative gradient scaled by the learning rate. To accurately evaluate model performance before deployment, practitioners employ k-fold cross-validation. The dataset is split into k equal partitions; the model trains on k-1 subsets and validates on the remaining fold, repeating k times to compute a robust average metric.

Lecture Notes: Databases and SQL Architecture

1. The Relational Model and Schemas

The relational database model organizes data into two-dimensional tables termed relations, composed of rows (tuples) and columns (attributes). Each tuple represents a single record identified uniquely by a primary key, while foreign keys enforce referential integrity across relational boundaries. Schemas specify table constraints, data types, and default values to prevent structural corruption.

2. ACID Properties of Transactions

Database transaction management relies on four fundamental ACID properties to guarantee reliability. Atomicity ensures that all operations in a transaction execute completely or roll back entirely with no partial state. Consistency guarantees that database state transitions preserve all defined schema rules and integrity constraints. Isolation dictates that concurrent transaction executions do not interfere with one another, preventing dirty reads through concurrency locks. Durability guarantees that once a transaction commits, its modifications persist permanently even across power failures.

3. Indexing Structures and B-Trees

Database indexes dramatically accelerate query search speeds by transforming full table scans of O(N) complexity into logarithmic time lookups of O(log N). The most prevalent indexing data structure is the balanced B-Tree (or B+ Tree), which maintains sorted keys across balanced hierarchical nodes, providing stable performance for both exact equality matches and range-based queries.

4. Normalization and SQL Relational Joins

Database normalization decomposes relations to eliminate insertion, update, and deletion anomalies while reducing duplicate storage. First Normal Form (1NF) demands atomic attribute values; Second Normal Form (2NF) eliminates partial key functional dependencies; and Third Normal Form (3NF) removes transitive dependencies. In SQL queries, relational joins recombine partitioned tables: an INNER JOIN produces only records matching both join conditions, whereas a LEFT OUTER JOIN preserves all rows from the left table supplemented with null values when no matching right records exist.

Lecture Notes: Operating System Process Scheduling

1. Process Lifecycle and Process Control Blocks

A process is a program in active execution requiring memory space, registers, and system resources. An operating system tracks each process across distinct lifecycle states: New during creation, Ready waiting in line for CPU assignment, Running executing instructions, Waiting or Blocked suspended pending I/O operations, and Terminated upon completion. The kernel maintains process state using a dedicated Process Control Block (PCB) containing the program counter, CPU registers, memory limits, and open file descriptors.

2. CPU Scheduling Algorithms

The CPU scheduler determines which ready process claims the processor next to maximize throughput and minimize response latency. First-Come First-Served (FCFS) schedules processes chronologically but suffers from the convoy effect when lengthy tasks delay brief jobs. Shortest Job First (SJF) minimizes average waiting time but risks starvation for elongated computational jobs. Round Robin (RR) scheduling allocates a fixed duration called a time quantum to each ready process in cyclic succession, ensuring preemptive fairness ideal for interactive time-sharing systems.

3. Context Switching Overhead

A context switch occurs when the OS kernel suspends one active thread and loads another. The kernel saves CPU registers, pointer addresses, and execution flags into the outgoing PCB and restores equivalent registers from the incoming PCB. Because context switching consumes CPU cycles without advancing user computation, time quanta in Round Robin must be tuned carefully to avoid excessive switching overhead.

4. Deadlocks and Virtual Memory Paging

A deadlock occurs when multiple processes are permanently halted because each holds a resource needed by another. Coffman identified four simultaneous conditions necessary for deadlock: Mutual Exclusion (non-shareable resources), Hold and Wait (processes holding allocated resources while requesting more), No Preemption (resources cannot be forcibly taken), and Circular Wait (a closed loop of waiting processes). In modern virtual memory architectures, logical address space is divided into fixed-size chunks called pages mapped to physical frames. When a process accesses a page not resident in RAM, a hardware interrupt triggers a page fault, prompting the OS to retrieve the page from disk via LRU replacement.

Lecture Notes: Computer Networks and Architecture

1. The OSI Seven-Layer Reference Model

The Open Systems Interconnection (OSI) model standardizes network communication into seven hierarchical abstraction layers. The Physical Layer transmits raw bit streams over hardware media; the Data Link Layer manages node-to-node framing and MAC addressing; the Network Layer directs packets through routing algorithms and IP addressing; the Transport Layer ensures end-to-end reliability and flow control; the Session Layer establishes dialog connections; the Presentation Layer handles data encryption and compression; and the Application Layer provides protocols directly for user software services.

2. Transport Protocols and the TCP Three-Way Handshake

The transport layer is dominated by TCP and UDP protocols. TCP (Transmission Control Protocol) is connection-oriented, delivering guaranteed, in-order packet streams through flow control, congestion avoidance, and sequence acknowledgments. Connection initiation requires a three-way handshake: the client transmits a SYN segment with an initial sequence number; the server responds with a SYN-ACK acknowledging the client's SYN; and the client replies with an ACK, establishing the bidirectional connection. In contrast, UDP (User Datagram Protocol) is connectionless and lightweight, sacrificing packet delivery guarantees to achieve low latency in real-time gaming, VoIP, and video streaming.

3. IP Addressing and Subnet Routing

Network devices are addressed via IPv4 (32-bit addresses formatted as dotted-decimal notation) or IPv6 (128-bit hexadecimal strings). Subnet masks delineate network prefixes from individual host portions, allowing hierarchical routing across autonomous systems. Routers employ routing protocols such as OSPF for interior gateway routing and BGP (Border Gateway Protocol) for exchanging reachability across the global Internet backbone.

4. DNS Resolution and Web Application Protocols

The Domain Name System (DNS) translates human-readable domain names into numerical IP addresses. When a user requests a URL, a recursive resolver queries Root nameservers, followed by Top-Level Domain (TLD) servers (e.g., .com or .org), and finally Authoritative DNS servers holding the definitive A or AAAA records. Web traffic communicates over HTTP and HTTPS; HTTPS secures plain-text HTTP by encrypting payloads using Transport Layer Security (TLS), preventing eavesdropping and man-in-the-middle tampering through public-key cryptography and digital certificates.